

Unit 3 Classical Encryption Techniques

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1.1 Symmetric-Key Encryption

* Definition: In symmetric-key encryption is a cryptographic technique in which the same secret key is used for both encryption and decryption

→ encryption → converting plaintext into ciphertext

→ decryption → converting ciphertext into plaintext

→ Sender and receiver must share the same secret key.

* Symmetric Cipher model:

→ Symmetric Cipher model describes how a message is encrypted and decrypted using a same shared secret key.

→ Formula:

$$\text{Encryption } C = E(K, P)$$

$$\text{Decryption } P = D(K, C)$$

where C = Cipher

E = Encryption

K = secret key

P = Plaintext

D = Decryption

* Model:

Plaintext



Encryption Algo + Key



Ciphertext



Decryption Algo + same Key



Plaintext

→ Example: HELLO

Key: 3 (Shift Key)

A	B	C	D	...
0	1	2	3	...

As above table

H = 7 → Shift this letter +3 position (Key=3) $= 7+3 = 10 = \text{K}$

E = 4 → +3 = 7 = H

L = 11 → +3 = 14 = O

L = 11 → +3 = 14 = O

O = 14 → +3 = 17 = R

Ciphertext = "KHOO R"

* Characteristics:

1. Uses one key for both encry. & decry.
2. Fast
3. Suitable for large data encryption
4. Faster than asymmetric encryption
5. Requires secure key sharing.

* Advantage:

- Fast
- Simple implementation
- less time

* Disadvantage:

- ^{very} Simple
- less secure
- Key distribution difficult

* Examples of Symmetric Algo:

- Caesar cipher
- Mono alphabetic
- Playfair
- Hill
- Vigenere

- Rail Fence

→ Modern Algo

- AES
- DES
- Triple DES
- Blowfish
- Twofish
- RC4

* 3.1.2 Cryptanalysis & Types of Attacks:

→ Cryptanalysis is the process of attempting to recover the plaintext or encryption key without knowing the secret key.

→ The person performing cryptanalysis is called "cryptanalyst".

→ Types of Attacks:

(1) Ciphertext-Only Attack (COA)

(2) Known-Plaintext Attack (KPA)

(3) Chosen-Plaintext Attack (CPA)

(4) Chosen-Ciphertext Attack (CCA)

(5) Chosen-Text Attack

- combination of chosen-plaintext & chosen-ciphertext attacks.

(6) Related-Key Attack

* Brute-Force Attack:

is a method of breaking encryption by trying every possible key until the correct one (key) is found.

steps:

- (1) Guess a key
- (2) Decrypt the message
- (3) Check output
- (4) Repeat until the correct key is found.

Example: If the key length is 8 bits:

$$\begin{aligned}\text{Total possible keys} &= 2^8 \\ &= 256 \text{ possible key}\end{aligned}$$

Try all 256 keys

Adv:

- Guaranteed to find the correct key if enough time is available.

- Does not require knowledge

→ No knowledge about encry/decr algo

Dis:

- Extremely time consuming for large key

- Modern algo with 128-bit / 256-bit keys are practically impossible to break the key.

(Replacement)

(Encryption)

3.2 Substitution Ciphers:

→ A substitution cipher is an ency. method in which each letter in the plaintext is replaced by another letter, number, or symbol according to a fixed rule.

→ Example

Plaintext: HELLO

Ciphertext: KHOOR (Key=3)

→ Types of Substitution ciphers

1) Monoalphabetic ciphers

2) Polyalphabetic

→ Alphabet & Numbers Table

A	B	C	D	E	F	G	H	I	J	K
0	1	2	3	4	5	6	7	8	9	10
L	M	N	O	P	Q	R	S	T	U	V
11	12	13	14	15	16	17	18	19	20	21
W	X	Y	Z							
22	23	24	25							

(single-letter)

1. Monoalphabetic Ciphers: (Encryption on Single-alphabet)

→ A Monoalphabetic Cipher is a type of substitution cipher in which each letter of the plaintext is replaced by one fixed letter of the ciphertext in entire message.

→ Once the rule (Key) is chosen, it remains the same for every letter.

→ Example: Plaintext: HELLO (Key=3)
Ciphertext: KHOOR

→ Types of Monoalphabetic Ciphers: (AM AM)
(the original trick)

- (1) Additive case (Caesar) Cipher
- (2) Multiplicative Cipher
- (3) Affine Cipher
- (4) Monoalphabetic Substitution Cipher

(1) Additive Cipher (Caesar Cipher)

→ The Additive Cipher, also called the Caesar Cipher, shifts every letter of the plaintext by a fixed number of positions (called the Key).

→ Encryption: $C = (P + K) \bmod 26$

Decryption: $P = (C - K) \bmod 26$

→ Example: HELLO (Key=3)

Ciphertext: KHOOR

→ Solution: Plaintext: HELLO (Key=3)

Plaintext	Key	$C = (P + K) \bmod 26$	Ciphertext
H = 7	3	$(7 + 3) \bmod 26 = 10$	K
E = 4	3	$(4 + 3) \bmod 26 = 7$	H
L = 11	3	$(11 + 3) \bmod 26 = 14$	O
L = 11	3	$(11 + 3) \bmod 26 = 14$	O
O = 14	3	$(14 + 3) \bmod 26 = 17$	R

(2) Multiplicative Cipher

→ In a Multiplicative Cipher, each plaintext letter is multiplied by a key before taking modulo 26.

→ Encryption $C = (P * K) \bmod 26$

Encryption $P = (C * K^{-1}) \bmod 26$

K^{-1} = multiplicative inverse of K

→ Valid Keys: The key must be coprime with 26.

Valid Key: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

→ Example: HELLO ($K=3$)

7 4 11 8 11 = 14

~> Encryption: Plaintext: HELLO (Key=3)

Plaintext	Key	$C = (P * K) \bmod 26$	Ciphertext
H = 7	3	$(7 * 3) \bmod 26 = 21$	V
E = 4	3	$(4 * 3) \bmod 26 = 12$	M
L = 11	3	$(11 * 3) \bmod 26 = 33 - 26 = 7$	H
L = 11	3	$(11 * 3) \bmod 26 = 7$	H
O = 14	3	$(14 * 3) \bmod 26 = 42 - 26 = 16$	@.

Ciphertext = VMHH@

(a) Affine Cipher:

→ The Affine Cipher is a combines multiplication and addition to encrypt the data.

→ Encryption $C = (aP + K) \bmod 26$

a = Multiplication Key

K = Shift Key

→ Decryption $P = a^{-1}(C - K) \bmod 26$

→ Example: HELLO ($K=8$) $a=5$, $K=8$

Plaintext	Letter number	$aP + K$	$(aP + K) \bmod 26$	Ciphertext
H	7	$5(7) + 8 = 43$	$43 \bmod 26$ $43 - 26 = 17$	R
E	4	$5(4) + 8 = 28$	$28 \bmod 26$ $28 - 26 = 2$	C
L	11	$5(11) + 8 = 63$	$63 \bmod 26$ $63 - 26 = 11$	L
L	11	$5(11) + 8 = 63$	$63 \bmod 26$ $63 - 26 = 11$	L
O	14	$5(14) + 8 = 78$	$78 \bmod 26$ $78 - 26 = 0$	A

Ciphertext = RCLLA

$$78 \div 26 = 3$$

$$3 \times 26 = 78$$

$$78 - 78 = 0$$

(4) Monoalphabetic Substitution Cipher:

→ In this method, each plaintext letter is replaced by another letter according to a randomly chosen substitution alphabet instead of using mathematical formulae.

→ Example Key:

Plain: A B C D E F G ...

Cipher: Q R K X Y U V ... A

→ Plaintext = CAT

C → K

A → Q

T → A

Ciphertext: KQA

Example:

Find $5!$

$5! = 1 \times 2 \times 3 \times 4 \times 5$

$= 120$

— x —

Find $3!$

$3! = 1 \times 2 \times 3$

$= 6$

→ Advantages:

- (1) Huge number of possible keys ($26!$) =
- (2) More secure than Caesar Cipher

→ Disadvantages:

- (1) Key distribution is difficult
- (2) Frequency analysis can still reveal the substitution pattern.

Comparison of Monoalphabetic Ciphers:

Cipher	Formula	Key	Security
Additive	$C = (P + K) \bmod 26$	Shift Key	Low
Multiplicative	$C = (P * K) \bmod 26$	Coprime with 26	Medium
Affine	$C = (aP + K) \bmod 26$	Two Keys (a, K)	Better
Monoalphabetic Substitution	— No Formula — (random alphabet)	26-letter mapping	Higher than Additive

(Can send any letters)

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Polyalphabetic Ciphers:

→ A Polyalphabetic Cipher is a substitution cipher in which multiple substitution alphabets are used to encrypt the plaintext.

→ Unlike monoalphabetic ciphers, the same plaintext letter may be encrypted into different ciphertext letters depending on its position or key.

→ Characteristics:

- (1) Uses multiple substitution alphabets.
- (2) Same plaintext letter can have different cipher letters.
- (3) More secure
- (4) Req. a keyword ^{and} ~~or~~ keys
- (5) Harder to break using frequency analysis

→ Adv:

- (1) More secure than monoalphabetic ciphers
- (2) Suitable for longer message encryption.

→ Disa:

- (1) More complex to implement.
- (2) Key distribution is more difficult.
- (3) Some methods can still be broken.

→ Types of Polyalphabetic Ciphers: (V, A, P, M-H)

(1) Vigenere Cipher (Rogier)

(2) Autokey "

(3) Playfair "

(4) Hill "

(5) One-Time Pad cipher

(1) Vigenere Cipher: (Caesar) (also called Vignere cipher)

→ The Vigenere Cipher is a cipher method that uses a repeating keyword for encryption.

→ It uses a Vigenere Table (Tabula Recta) consisting of 26 shifted alphabets (26x26 grid)

→ Encryption $C = (P + K) \bmod 26$

→ Decryption $P = (C - K) \bmod 26$

→ Example: HELLOWELCOME

Keyword: LEMON

Repeated Key

H	E	L	L	O	W	E	L	C	O	M	E
L	E	M	O	N	L	E	M	O	N	L	E

- Encryption: (Tabula Recta)

	A	B	C	D	E	F	G	H	I
A	A	B	C	D	E	F	G	H	I
B	B	C	D	E	F	G	H	I	J
C	C	D	E	F	G	H	I	J	K
D	D	E	F	G	H	I	J	K	L
E	E	F	G	H	I	J	K	L	M
F	F	G	H	I	J	K	L	M	N
G	G	H	I	J	K	L	M	N	O
H	H	I	J	K	L	M	N	O	P

Encryption:

Plaintext	Keyword	$(P+K) \bmod 26$	Ciphertext
H = 7	L = 11	$(7+11) \bmod 26 = 18$	S
E = 4	E = 4	$(4+4) \bmod 26 = 8$	I
L = 11	M = 12	$(11+12) \bmod 26 = 23$	X
L = 11	O = 14	$(11+14) \bmod 26 = 25$	Z
O = 14	N = 13	$(14+13) \bmod 26 = 1$	B
W = 22	L = 11	$(22+11) \bmod 26 = 7$	H
E = 4	E = 4	$(4+4) \bmod 26 = 8$	I
L = 11	M = 12	$(11+12) \bmod 26 = 23$	X
C = 2	O = 14	$(2+14) \bmod 26 = 16$	Q
O = 14	N = 13	$(14+13) \bmod 26 = 1$	B
M = 12	L = 11	$(12+11) \bmod 26 = 23$	X
E = 4	E = 4	$(4+4) \bmod 26 = 8$	I

Solution

$$(14+13) \bmod 26 = 27 \bmod 26$$

$$= 27 - 26$$

$$= 1$$

$$(22+11) \bmod 26 = 33 \bmod 26$$

$$= 33 - 26$$

$$= 7$$

Ciphertext = SIXZBXIXQBXI

(2) Autokay Cipher:

→ The Autokay Cipher is an improvement over the Vigenere cipher.

→ It uses a keyword followed by the plaintext itself to generate the complete key.

→ Working:

(1) Choose a keyword

(2) Write the keyword

(3) Append (copy/paste) the plaintext to the keyword until the key length equals the plaintext length.

(4) Encrypt using the Vigenere method.

→ Example:

Plaintext: HELLO

Keyword: KEY

Generated Key: KEY HE

→ Formula:

(नेसु)

$$\text{Ciphertext} = \text{Key} + \text{Plaintext}$$

using Vigenere method)

→ Encryption:

$$C = (P + K) \bmod 26$$

Plaintext	Key	$C = (P + K) \bmod 26$	Ciphertext
H = 7	K = 10	$(7 + 10) \bmod 26 = 17$	R
E = 4	E = 4	$(4 + 4) \bmod 26 = 8$	I
L = 11	Y = 24	$(11 + 24) \bmod 26 = 9$	J
L = 11	H = 7	$(11 + 7) \bmod 26 = 18$	S
O = 14	E = 4	$(14 + 4) \bmod 26 = 18$	S

Solution

$$\begin{aligned}
 &= (11 + 24) \bmod 26 \\
 &= 35 \bmod 26 \\
 &= 35 - 26 \\
 &= 9
 \end{aligned}$$

Encryption $C = (P + K) \bmod 26$

Decryption $P = (C - K) \bmod 26$

Ciphertext = R I J S S

→ Decryption $P = (C - K) \bmod 26$

Ciphertext	Key	$(C - K) \bmod 26$	Plaintext
R = 17	K = 10	$(17 - 10) \bmod 26 = 7$	H
I = 8	E = 4	$(8 - 4) \bmod 26 = 4$	E
J = 9	Y = 24	$(9 - 24) \bmod 26 = 11$	L
S = 18	H = 7	$(18 - 7) \bmod 26 = 11$	L
S = 18	E = 4	$(18 - 4) \bmod 26 = 14$	O

Plaintext = "HELLO"

$$\begin{aligned}
 &= (9 - 24) \bmod 26 \\
 &= 15 \bmod 26 \\
 &= 26 - 15 \\
 &= 11
 \end{aligned}$$

(3) Playfair Cipher:

- The Playfair Cipher encrypts pairs of letters (digraphs) instead of individual letters.
- It uses 5x5 matrix with a keyword.
- The letters I and J are treated as one letter.
- X is added if necessary.
- The Playfair cipher is a manual symmetric encryption technique that encodes pairs of letters (digrams) instead of single characters.
- Invented in 1854 by Charles Wheatstone and promoted by Lord Playfair.

→ Encryption Rules

- (1) Same-Row: If both letters are in same row, replace each with the letter to its right.
- (2) Same-Column: If both letters are in same-column, replace each with the letter below it.
- (3) Different Row & Column: Otherwise form a rectangle and replace each letter with the letter in the same row but the other letter's column.

* Advantages:

- (1) Encrypts two letters at a time.
- (2) Better security.
- (3) Hides single-letter frequencies.

5x5 matrix

A	B	C	D	E
F	G	H	I/J	K
L	M	N	O	P
Q	R	S	T	U
V	W	X	Y	Z

* Disadvantages:

- (1) More complex.
- (2) Repeated digraphs may still reveal patterns.

* How to create 5x5 Matrix:

Example: Keyword = MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

* સાચેથી Keyword 'MONARCHY' ને 5x5 matrix માં લેવામાં આવે છે.

* જો Keyword માં કોઈ character repeat થતો હોય તો બિગ વાર તેને matrix માં દાખલ નથી કરવામાં આવે.

* Matrixમાં Keyword લખ્યા પછી (A-Z) character દાખલ કરવામાં આવે છે જેથી Keyword લખ્યા પછી બાકીના character લેવામાં આવે છે.

* I & J ને સાચે સાચે રાખવામાં આવે છે.

Example: Encrypt the message
 "Hide the gold under the carpet" using
 'play fair' technique with the keyword
 "Neso Academy".

Plaintext: "Hide the gold under the carpet"

Keyword: "Neso Academy".

* In 2-character pair mein last me ek character
 hoga "X" dena hoga.

Pairs of letters: ^{extra} (X added)

Plaintext:	H	i	d	e	t	h	e	g	o	l	d	u	n	d	e	r	t	h	e	c	a	r	p	e	t
Ciphertext:	I	K	G	D	A	K	D	P	N	R	C	V	E	C	O	P	A	K	N	D	O	T	X		

Solution: 5x5 Matrix

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Same-row
 Right-side)

Hi → IK
 I → K

using * de

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(same-column-Rule)
below-letter

~~d~~ → G
~~e~~ → D

* th

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Rectangle rule)

P	E	T	X
V	D	R	Z

t → Q
h → K

* eg

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(column-rule)
below-letter

e → D
g → P

* OL (Rectangle-Rule)

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

O → N
L → R

* DU (Rectangle-Rule)

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

D → C
U → V

* ND (Rectangle-Rule)

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

N → E
D → C

* ER (Rectangle-Rule)

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

E → O
R → P

*TH

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Rectangle-Rule)

T ~~~> Q

H ~~~> K

*EC

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Rectangle-Rule)

E ~~~> N

C ~~~> D

*AR

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Rectangle-Rule)

A ~~~> O

R ~~~> T

*PE

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

(Same-Column-Rule)
select-down-letter

P ~~~> V

E ~~~> D

* TX	N	E	S	O	A	(Rectangle-Rule)
	C	D	M	Y	B	
	F	G	H	I/J	K	T $\rightsquigarrow$ A
	L	P	Q	R	T	X $\rightsquigarrow$ Z
	U	V	W	X	Z	

Final

Ciphertext = IKGDQKDPNR/VECPQKND
OTVDRZ"

→ **Example: 2** Encrypt the message "ATTACK" using Playfair technique with the keyword "MONARCHY".

⑥ Pair of Letters:

Plain text	A	T	C
Cipher text	S	N	K

© Matrix:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

 $A \subseteq R$
$$T = S$$

55

$$T = S$$

Ciphertext = "RSSRDE"

$$R = \varepsilon$$

4) Hill Cipher:

- The Hill Cipher is a polyalphabetic cipher that uses matrix multiplication for encryption.
- This method encrypt blocks of letters, instead of one letter at a time.
- The Hill Cipher is a polygraphic substitution block cipher based on linear algebra.
- It converts blocks of plaintext into blocks of ciphertext using a square key-matrix.
- American mathematician Lester S. Hill developed this technique in 1929.
- Formula:

Encryption: $C = (K \cdot P) \text{ mod } 26$

Here $K = n \times n$ Key matrix

Decryption: $P = (K^{-1} \cdot C) \text{ mod } 26$

Here K^{-1} = inverse of the Key matrix modulo 26

→ Working:

1. Convert letters into numbers
2. Divide plaintext into blocks
3. Multiply each block by the key-matrix
4. Take modulo 26
5. Convert numbers back into letters (cipher).

* Adv:

- (1) Encrypts multiple letters together.
- (2) Strong algo.
- (3) Resistant to simple frequency analysis.

* Dis:

- (1) Complex
- (2) Matrix inverse is required for decryption.
- (3) Invalid key-matrix can't be used.

Example I Plaintext = "HI" with Key $\begin{bmatrix} 6 & 24 \\ 1 & 13 \end{bmatrix}$

$$\begin{bmatrix} H \\ I \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \quad K = \begin{bmatrix} 6 & 24 \\ 1 & 13 \end{bmatrix}$$

$$K \cdot P = \begin{bmatrix} 7 & 8 \end{bmatrix} \times \begin{bmatrix} 6 & 24 \\ 1 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} (7 \times 6) + (8 \times 24) \\ (7 \times 1) + (8 \times 13) \end{bmatrix}$$

$$= \begin{bmatrix} 42 + 192 \\ 7 + 104 \end{bmatrix} = \begin{bmatrix} 234 \\ 111 \end{bmatrix}$$

* Apply modulo 26

$$\begin{aligned} &= (234) \bmod 26 = 11 \\ &= (26 \times 9) + 0 \end{aligned}$$

$$= 0$$

$$\begin{aligned} &= 11 \bmod 26 \\ &= (26 \times 4) + 7 \end{aligned}$$

$$= 7$$

$$= \begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} A \\ H \end{bmatrix}$$

Ciphertext = "AH"

Example: 2 Encrypt the plaintext "HI" using Hill Cipher. (Key-matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$)

create plaintext matrix

step 1: $H = 7$

$I = 8$

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$C = K \cdot P \pmod{26}$$

step 2: (multiply plaintext matrix * Key-matrix)

$$K \cdot P = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \times \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} (7 \times 3) + (8 \times 2) \\ (7 \times 3) + (8 \times 5) \end{bmatrix}$$

$$= \begin{bmatrix} 21 + 16 \\ 21 + 40 \end{bmatrix} = \begin{bmatrix} 37 \\ 61 \end{bmatrix}$$

step 3: Apply modulo 26

$$37 \pmod{26} = 37 - 26 = 11$$

$$61 \pmod{26} = (26 \times 2) + 9 = 9$$

$$C = \begin{bmatrix} 11 \\ 9 \end{bmatrix}$$

step 4: Convert numbers into letters

$$11 = T$$

$$9 = C$$

Ciphertext = "TC"

Example: 3

Plaintext: ACT

Key: GYBNQKURP

Step 1: Create plaintext matrix = ACT

$n=3$

$P = \begin{bmatrix} A & C & T \\ 0 & 2 & 19 \end{bmatrix}$ (Three characters)

Key-matrix $(3 \times 3) = \begin{bmatrix} G & Y & B \\ N & Q & K \\ U & R & P \end{bmatrix}$
(because $n=3$)

Step 2: Multiplication = $K \cdot P$

$K \cdot P = \begin{bmatrix} 0 & 6 & 24 & 1 \\ 2 & 13 & 16 & 10 \\ 19 & 20 & 17 & 15 \end{bmatrix}$

$= \begin{bmatrix} (0 \times 6) + (2 \times 24) + (19 \times 1) \\ (0 \times 13) + (2 \times 16) + (19 \times 10) \\ (0 \times 20) + (2 \times 17) + (19 \times 15) \end{bmatrix} = \begin{bmatrix} 0 + 48 + 19 \\ 0 + 32 + 190 \\ 0 + 34 + 285 \end{bmatrix}$

$= \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix}$

step 4: Apply(modulo 26)

$= 67 \text{ modulo } 26$ $= (26 \times 2) + 15 = 15$ $= 52 + 15$ $= \textcircled{15}$	$= 222 \text{ mod } 26$ $= (26 \times 8) + 14$ $= \textcircled{14}$	$= 319 \text{ mod } 26$ $= (26 \times 12) + 7$ $= \textcircled{7}$
--	---	--

15
14
7

Step 5: Convert numbers into letters

$$15 = P$$

$$14 = O$$

$$7 = H$$

Ciphertext = POH

Matrix Multiplication

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad B = \begin{bmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{bmatrix}$$

$$A * B = \begin{bmatrix} a_1 & * & \begin{bmatrix} b_1 & c_1 & d_1 \end{bmatrix} \\ a_2 & * & \begin{bmatrix} b_2 & c_2 & d_2 \end{bmatrix} \\ a_3 & * & \begin{bmatrix} b_3 & c_3 & d_3 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} (a_1 * b_1) + (a_2 * c_1) + (a_3 * d_1) \\ (a_1 * b_2) + (a_2 * c_2) + (a_3 * d_2) \\ (a_1 * b_3) + (a_2 * c_3) + (a_3 * d_3) \end{bmatrix}$$

2x2 matrix:

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \quad B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$$

$$A * B = \begin{bmatrix} (a_1 * b_1) + (a_2 * b_3) & (a_1 * b_2) + (a_2 * b_4) \\ (a_3 * b_1) + (a_4 * b_3) & (a_3 * b_2) + (a_4 * b_4) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 & * & \begin{bmatrix} b & c \end{bmatrix} \\ a_2 & * & \begin{bmatrix} d & e \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} a_1 b + a_2 c \\ a_1 d + a_2 e \end{bmatrix}$$

Example 4 Plaintext = CODE

Key matrix = Keyword = "DCDF"

Step 1: Create matrix:

Key-matrix

Plaintext = CODE

Keyword = DCDF

$$P = \begin{vmatrix} C & D \\ O & E \end{vmatrix}$$

Here Keyword matrix can be create (2x2)

$$= \begin{vmatrix} D & D \\ C & F \end{vmatrix}$$

→ Now, Matrix multiplication needs a vector of size (2x1),

So, we can break plaintext matrix into two-letter blocks

$$\begin{vmatrix} C \\ O \end{vmatrix} \text{ and } \begin{vmatrix} D \\ E \end{vmatrix}$$

Step 2: Multiplication (K.P)

$$\begin{vmatrix} C \\ O \end{vmatrix} * \begin{vmatrix} D & D \\ C & F \end{vmatrix} \rightarrow \begin{vmatrix} 33 & 2 \\ 25 & 14 \end{vmatrix} =$$

$$\begin{vmatrix} D \\ E \end{vmatrix} * \begin{vmatrix} D & D \\ C & F \end{vmatrix} \rightarrow \begin{vmatrix} 21 & 33 \\ 26 & 25 \end{vmatrix}$$

$$\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 14 & 25 \\ \hline \end{array} * \begin{array}{|c|c|} \hline 3 & 3 \\ \hline 2 & 5 \\ \hline \end{array} = \begin{array}{|c|c|} \hline (2*3) + (14*3) & \\ \hline (2*2) + 14*5 & \\ \hline \end{array} = \begin{array}{|c|c|} \hline 6+42 & \\ \hline 4+70 & \\ \hline \end{array} = \begin{array}{|c|c|} \hline 48 & \\ \hline 74 & \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline 3 & 3 \\ \hline 4 & 25 \\ \hline \end{array} * \begin{array}{|c|c|} \hline 3 & 3 \\ \hline 2 & 5 \\ \hline \end{array} = \begin{array}{|c|c|} \hline (3*3) + 3*4 & \\ \hline 3*2 + (4*5) & \\ \hline \end{array} = \begin{array}{|c|c|} \hline 9+12 & \\ \hline 6+20 & \\ \hline \end{array} = \begin{array}{|c|c|} \hline 21 & \\ \hline 26 & \\ \hline \end{array}$$

Step 4 Apply modulo

$$48 \bmod 26 = 48 - 26 = 22$$

$$74 \bmod 26 = (26*2) + 22 = 22$$

$$21 \bmod 26 = 21$$

$$26 \bmod 26 = 0$$

$$= \begin{array}{|c|c|} \hline 48 & 21 \\ \hline 74 & 26 \\ \hline \end{array} \bmod 26$$

$$= \begin{array}{|c|c|} \hline 22 & 21 \\ \hline 22 & 0 \\ \hline \end{array}$$

Step 5 Convert numbers into letters

22 22 21 0

W W V A $\Rightarrow$ Cipher text = "WWVA"

(5) One-Time Pad (OTP)

- The OTP is the most secure polyalphabetic cipher.
- It uses completely random key that is;
 - Random
 - Equal in length to the plaintext.
 - Used only once
- OTP also known as the "Vernam cipher".
- This method is mathematically proven to be unbreakable.
- The key must be used strictly once and then destroyed.

→ Encryption $C = (P + K) \bmod 26$

→ Encryption $P = (C - K) \bmod 26$

→ Advantages:

- (1) Perfect secrecy.
- (2) Impossible to break
- (3) Immune to brute-force and frequency analysis

→ Disadvantages:

- (1) Key must be truly random.
- (2) Key length must equal to the message length
- (3) Key distribution is difficult.
- (4) Key can't be reused.

→ Note: In OTP method → "Message length = Key length"

→ Example: Plaintext: "HELLO"

secret Key: "XMCKL"

→ Encryption $C = (P + K) \bmod 26$

Plaintext	Key	$(P + K) \bmod 26$	Ciphertext
H = 7	X = 23	$(7 + 23) \bmod 26 = 30 \bmod 26 = 30 - 26 = 4$	E
E = 4	M = 12	$(4 + 12) \bmod 26 = 16 \bmod 26 = 16$	Q
L = 11	C = 2	$(11 + 2) \bmod 26 = 13$	N
L = 11	K = 10	$(11 + 10) \bmod 26 = 21$	V
O = 14	L = 11	$(14 + 11) \bmod 26 = 25$	Z

→ Ciphertext = "EQNVZ"

→ OTP was first described in 1882 by American banker Frank Miller.

→ Later, patented as an automated teleprinter cipher in 1917 by Gilbert Vernam of AT&T (Bell labs).

→ In 1949, mathematician Claude Shannon published research proving mathematically that the OTP truly provides "perfect secrecy" (unbreakable).

Decryption: Ciphertext: "EGENVZ"
 (OTP)

Secret Key: "XMCKL"

→ Formula $P = (C - K) \bmod 26$

Ciphertext	Key	$(C - K) \bmod 26$	Plaintext
E = 4	X = 23	$(4 - 23) = -19 \bmod 26 = -19 + 26 = 7$	H
G = 16	M = 12	$(16 - 12) = 4 \bmod 26 = 4$	E
N = 13	C = 2	$(13 - 2) = 11 \bmod 26 = 11$	L
V = 21	K = 10	$(21 - 10) = 11 \bmod 26 = 11$	L
Z = 25	L = 11	$(25 - 11) = 14 \bmod 26 = 14$	O

→ Plaintext = "HELLO"

Comparison of Polyalphabetic Ciphers:

Cipher	Key Used	Unit of Encryption	Security
Vigenere	Repeating Keyword	Single letter	High
Autokey	Keyword + Plaintext	Single letter	High
Playfair	5x5 Matrix	Letter Pairs	High
OTP	Random Key Matrix Multiplication	Letter blocks	Very High
Hill	Matrix Multiplication Random Key	Single-letter	Very High

3.3 Transposition Ciphers:

→ A Transposition Cipher is an encryption method in which the positions of the plaintext letters are rearranged according to a specific rule or Key.

→ The letters of plaintext are not changed. It just rearranged.

→ Formula: Plaintext → Rearrange letters → Ciphertext

→ Decryption is done by placing the letters back into their original positions.

→ Characteristics:

- (1) Same letters appear in ciphertext.
- (2) Only positions change.
- (3) Easy to implement.
- (4) less secure

→ Example:

Plaintext: HELLO

Rearranged: OLLEH

Ciphertext: OLLEH

→ Type of Transposition:

(1) Keyless Transposition

(2) Keyed Transposition

(3) Stream & Block Ciphers

3.3.1 Keyless Transposition:

- In Keyless transposition, no secret key is used.
- The rearrangement follows a fixed pattern.
- There are two type of Keyless transposition.
 - A. Rail Fence Cipher
 - B. Rectangular (columnar) Transposition.

→ A. Rail Fence Cipher:

- ① In Rail Fence Cipher, writes the message diagonally across multiple rows (Rails) and then read it row by row.

② Encryption Steps:

- (1) Write the plaintext diagonally.
- (2) Move downward and upward alternately.
- (3) Read each row from left to right.

→ Example: Plaintext: HELLO WORLD

Using 2 rails: HELLO WORLD

H	L	O	W	R	D
E	L		O	L	

Ciphertext: HLOWRDELOL (without spaces)

→ Advantages:

- (1) Very Simple
- (2) Fast Encryption

→ Disadvantages:

- (1) Easy to break
- (2) Less secure

→ B. Rectangular (Columnar) Transposition:

- ① The plaintext is written by row inside a rectangle with fixed columns.
- ② The ciphertext is obtained by reading the columns (up-to-down) one after another.
- ③ Example:

Plaintext: ATTACK AT DAWN

Arrange into 4 columns.

A	T	T	A
C	K	A	T
D	A	W	N

Read column-wise (Top-to-down)

A C D T K A T A W A T N

Ciphertext: "ACDTRATAWATN"

④ Adv:

- 1) Easy to understand
- 2) Very simple
- 3) Better than Rail-Fence

④ Dis:

- 1) Pattern can be guessed.
- 2) Less secure

3.3.2 Keyed Transposition Method:

- In this method, a secret Keyword determines the order of columns.
- Only someone who knows the key can decrypt the message.
- In this method, rearranges the positions of characters in a message using a secret keyword or password or a secret numerical sequence.

→ Example: Plaintext: "MEET ME"

Secret Key: 3, 1, 4, 2

Block Size: 4 characters

Step: 1 Create 4-character block

Plaintext: [M][E][E][T] [M][E][X][X]

Block-1

Block-2

↑↑ Here x added if necessary

Step: 2 Take Block-1

Position 1: M (1st letter)

Position 2: E (2nd ")

Position 3: E (3rd ")

Position 4: T (4th ")

step: 3: Apply Key (3, 1, 4, 2)

New 1st Position = 3 $\Rightarrow$ E

New 2nd " = 1 $\Rightarrow$ M

New 3rd Position = 4 $\Rightarrow$ T

New 4th Position = 2 $\Rightarrow$ E

Result for Block-1 = "EMTE"

step: 4: Take Block-2

M E X X

Position 1: M

" 2: E

" 3: X

" 4: X

Now Apply Key (3, 1, 4, 2)

New 1st Position = 3 $\Rightarrow$ X

New 2nd " = 1 $\Rightarrow$ M

New 3rd " = 4 $\Rightarrow$ X

New 4th " = 2 $\Rightarrow$ E

Result for Block-2 = "XMXE"

Final Ciphertext = "EMTEXMXE"

3.3.3 Stream Cipher & Block Ciphers:

→ Stream cipher and Block Ciphers are two important methods used in cryptography to encrypt data.

→ Both are symmetric-key encryption methods.

→ Meaning the same-key is used for both encryption & decryption.

→ Stream Cipher: XOR

① A Stream Cipher encrypts plaintext one bit (one-character) at a time.

② Each plaintext bit is combined with a key stream to produce ciphertext.

③ Typically using an exclusive-or (XOR) operation. (bit-by-bit OR byte-by-byte)

④ XOR operation:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0$$

⑤ Working:

$$\text{Plaintext} \oplus \text{Keystream} \Rightarrow \text{Ciphertext}$$

(XOR)

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② Example: Plaintext: 1100101 (in binary)

Keystream: 0100111 (in binary)

* Calculation: Plaintext $\oplus$ Keystream

1100101
(XOR) $\oplus$ 0100111
1000010

Ciphertext = 1000010

③ Characteristics:

- (1) Encrypts data continuously.
- (2) Processes one bit/byte at a time
- (3) Requires Keystream
- (4) Use XOR operation

④ Advantages:

- (1) High encryption speed.
- (2) Requires less memory.
- (3) less time requires for encryption
- (4) suitable for real-time communication.

⑤ Disadvantages:

- (1) Reusing the same key
- (2) less secure

⑥ Applications: (a) - Wireless communication

- (2) Voice calls
- (3) Video streaming
- (4) online chat application.

→ Block Cipher:

①: A Block Cipher encrypts data in fixed-size blocks (like 64-bits or 128-bits) instead of one bit at a time.

② A Block cipher is a symmetric-key cryptography method that encrypts data in fixed-size blocks using a secret key.

→ ③ Examples (Here we take just tiny 4-bit example)

Plaintext: 1101

Secret Key: ~~1101~~

(Rule)

Shift each bit position forward by 1 value

1011

↓↓↓↓

0100

Ciphertext: 0100

④ Steps:

① Split Block

② Apply Substitution

③ Apply XOR operation

Example: 2 (Tiny 4-bit Block Cipher)

Block Size: 4-bit

Key: 1101

Plaintext: 1011

Step: 1 - Split Block: Break 4-bit plaintext into two 2-bit values:

$\boxed{10} \quad \boxed{11}$

Step: 2 Apply Substitution:

$\begin{array}{r} 0 \ 1 \\ 1 \ 0 \\ - \ 1 \\ \hline 0 \ 0 \end{array}$	$\begin{array}{r} 1 \ 1 \\ - \ 1 \\ \hline 1 \ 0 \end{array}$
--	---

Now after substitution,

block = $\boxed{0010}$ (4-bit)

Step: 3 Apply XOR operation with Key

Substitution block: 0010
 Key: $\oplus$ 1101
 (XOR) = 1111

$\boxed{\text{Ciphertext} = 1111}$

⑤ Common block cipher modes:

- (1) ECB: Electronic Code book
- (2) CBC: Cipher Block Chaining
- (3) CFB: Cipher Feedback Mode
- (4) OFB: Output Feedback Mode
- (5) CTR: Counter Mode

⑥ Characteristics:

- (1) Encrypts fixed-size blocks.
- (2) More secure than stream-ciphers.
- (3) Widely used in modern cryptography.

⑦ Advantages:

- (1) Strong security.
- (2) Suitable for large files.
- (3) Supports different modes of operation.

⑧ Disadvantages:

- (1) Slower than stream-ciphers
- (2) Requires more memory.
- (3) Padding may be needed.

⑨ Applications:

- (1) File encryption
- (2) Disk "
- (3) Database security
- (4) HTTPS and secure communication.

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→ Difference bet^w Stream Cipher & Block Cipher:

Stream Cipher	Block Cipher
Encrypts one bit/byte at a time	Encrypts fixed-size blocks
Faster	Slower
Requires keystream	Uses encryption rounds
No padding required	Padding may be required.
Less memory requirement	Requires more memory
Suitable for real-time data	Suitable for stored data

3.4 Asymmetric-Key Cryptosystem : (Public-Key Cryptography)

① An Asymmetric-Key cryptosystem is an encryption system that uses two-different keys.

② Public Key: Used for encryption and can be shared with everyone.

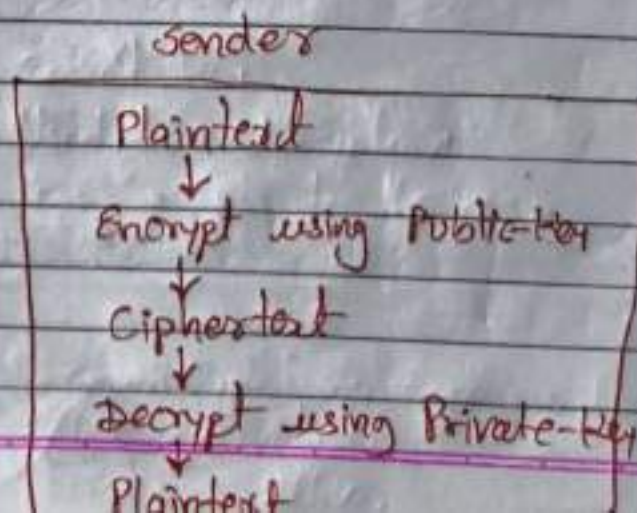
③ Private Key: Used for decryption and is kept secret by the owner.

④ Unlike symmetric-key encryption, the sender and receiver do not share the same key.

⑤ Working Principle:

1. The receiver generates a public key and private key.
2. The public key is shared openly.
3. The sender encrypts plaintext using receiver's public key.
4. The encrypted message is sent.
5. The receiver decrypts the message (ciphertext) using the private-key.

⑥ Diagram:



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⑥ Features:

- (1) Uses two different keys.
- (2) Public key is openly distributed.
- (3) Private key is kept secret.
- (4) Solves the key-distribution problem.
- (5) Provides secure communication.

⑦ Advantages:

- (1) High security.
- (2) No need to exchange secret keys.
- (3) Supports digital signature.
- (4) Provides authentication & confidentiality.

⑧ Disadvantages:

- (1) Slower than symmetric-key encryption.
- (2) Requires more power.
- (3) More complex & implementation

⑨ Applications:

- (1) Secure web browsing (HTTPS)
- (2) Email encryption.
- (3) Digital signatures
- (4) Online banking.
- (5) E-commerce
- (6) VPNs.

→ 3.4.1 Ingredients of Public-Key Encryption:

→ A public-key cryptosystem consists of the following components:

- (1) Plaintext
- (2) Encryption Algorithm
- (3) Public-Key
- (4) Private-Key
- (5) Ciphertext
- (6) Decryption Algo.

→ 1. Plaintext: The original readable message.
Example: HELLO

→ 2. Encryption Algo: The mathematical algo that converts plaintext into ciphertext using the public-key.

→ 3. Public-Key:

- Shared openly.
- Used only for encryption
- Anyone can use it.

→ 4. Private-Key:

- Secret Key
- Used for decryption
- Known only to the owner

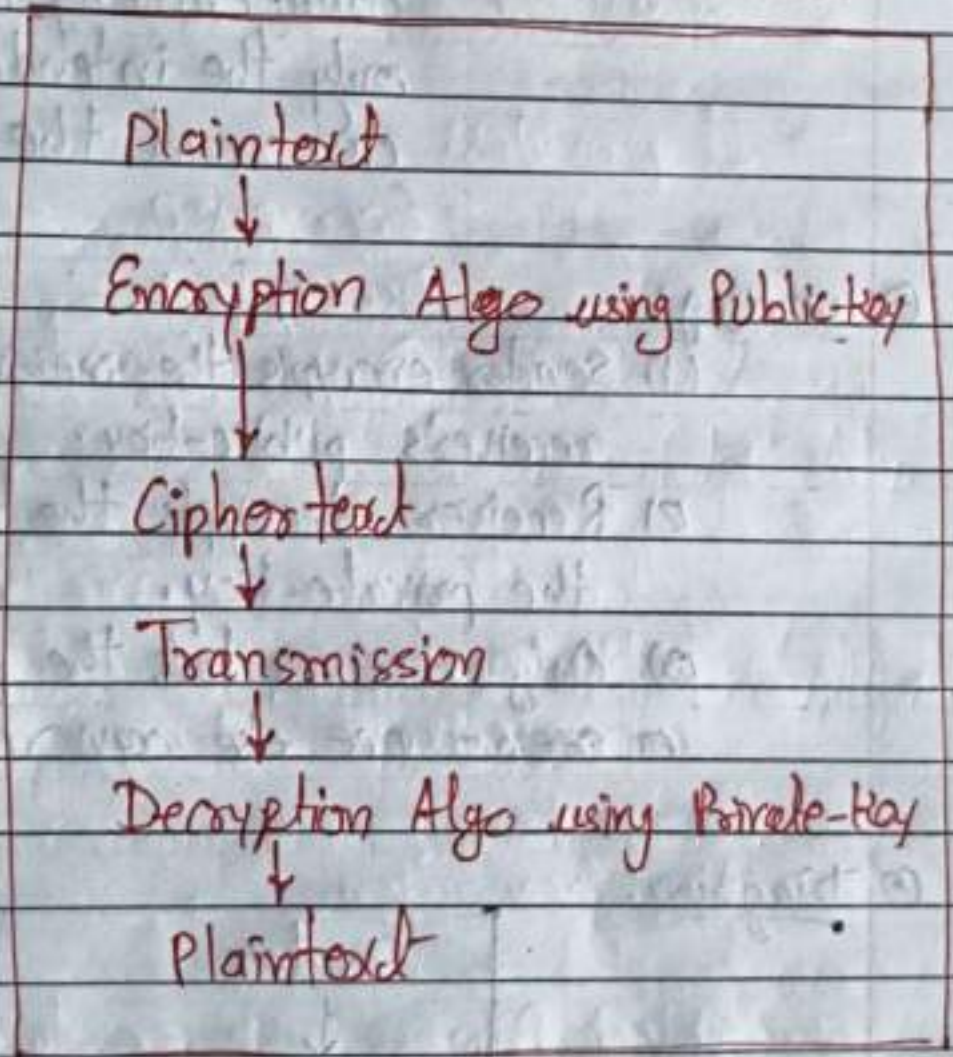
→ 5. Ciphertext:

- The encrypted message produced after encryption.
- It is unreadable.

→ Decryption Algo:

The algo that converts ciphertext back into plaintext (original message) using the private-key.

→ Diagram:



3.4.2 Confidentiality & Authentication in Public-Key Cryptography:

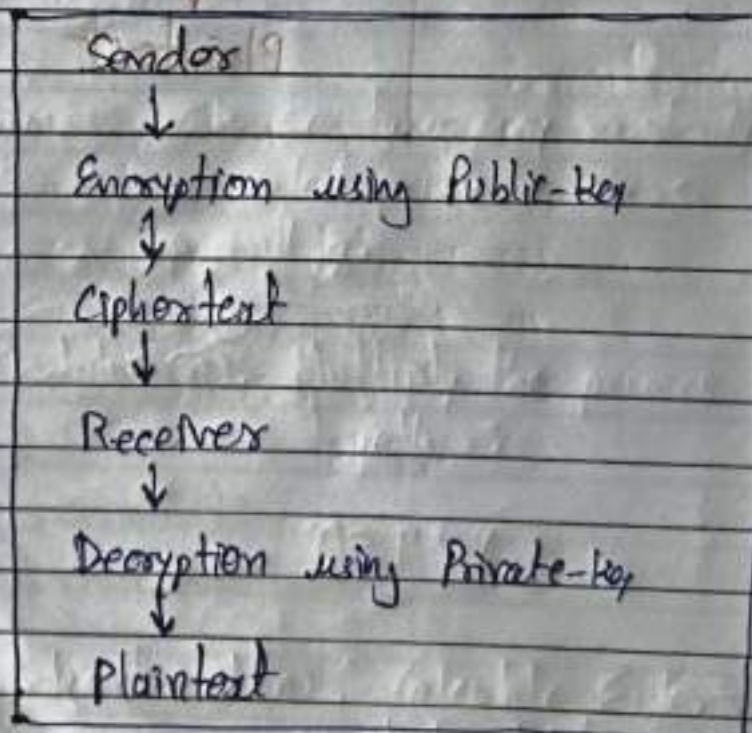
① Confidentiality:

Confidentiality ensures that only the intended receiver can read the message.

② Working:

- (1) Sender encrypts the message using the receiver's public-key.
- (2) Receiver decrypts the message using the private-key.
- (3) Only receiver has the private-key, ~~no~~ ~~sender~~ no one else can read the message.

③ Diagram:



- #### ④ Advantages:
- (1) Prevents unauthorized access.
 - (2) Protects sensitive information.
 - (3) Ensures privacy.

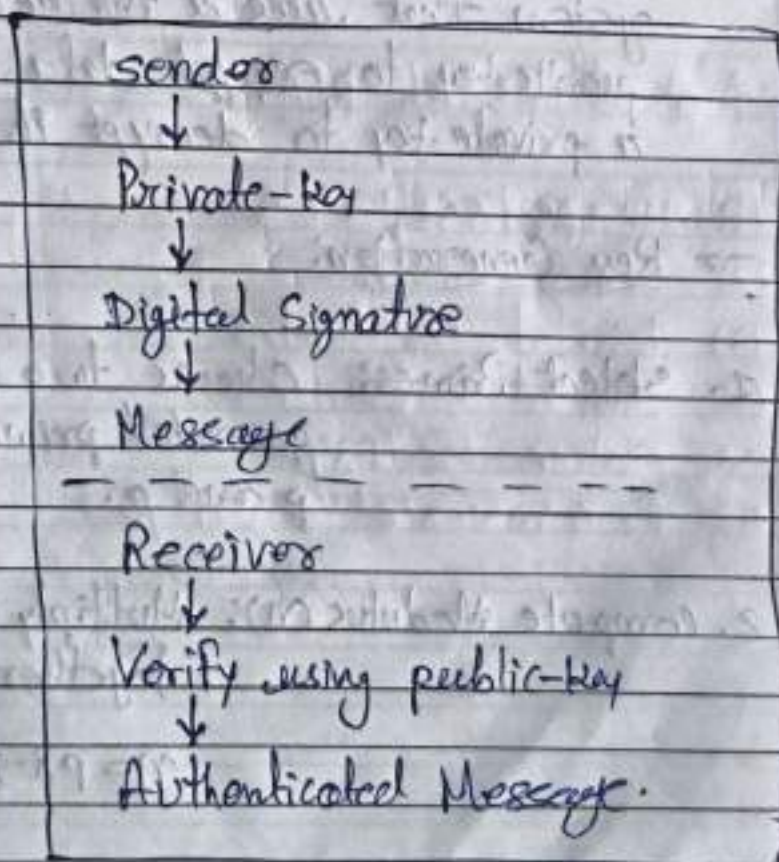
① Authentication:

Authentication verifies the identity of the sender and confirms that the message has not been modified.

② Working:

- (1) Sender creates a digital signature using their private-key.
- (2) Receiver verifies the signature using the sender's public-key.
- (3) If verification succeeds, the sender is authentic.

③ Diagram:



④ Advantages:

- (1) Verifies sender identity.
- (2) Ensures data integrity.
- (3) Provides non-repudiation.
- (4) Secure communication.

3.4.3 Basic Working of RSA Algorithm:

→ RSA: Rivest-Shamir-Adleman

→ RSA is one of the most widely used public-key cryptographic algorithms.

→ It provides:

- Encryption
- Decryption
- Digital Signature.

→ The RSA algo is an asymmetric cryptographic system that uses a mathematically linked public-key to encrypt data and a private-key to decrypt it.

→ Key Generation:

1. Select Primes: Choose two any randomly generated prime numbers.
 p and q .

2. Compute Modulus (n): Multiply the primes together:

$$n = p \times q$$

3. Compute Totient $\phi(n)$: Calculate Euler's totient function

$$\phi(n) = (p-1) \times (q-1)$$

4. Choose Public Exponent (e):

select an integer (e) that is coprime with $\phi(n)$, meaning $\gcd(e, \phi(n)) = 1$

5. Compute Private Exponent (d):

calculate (d) as the modular multiplicative inverse of (e) modulo $\phi(n)$.

$$\begin{cases} d \times e \equiv 1 \pmod{\phi(n)} \\ (d \times e) \bmod \phi(n) \equiv 1 \end{cases}$$

6. Resulting Keys:

⊙ Public-Key: Shared with anyone, pair (e, n):

⊙ Private-Key: Kept secret by the receiver, pair (d, n):

→ RSA Encryption Formula:

$$C = m^e \pmod{n}$$

where, m = Plaintext

e = Public Key

n = modulus

C = Ciphertext

d = Private Key

→ RSA Decryption Formula:

$$m = C^d \pmod{n}$$

→ Advantages of RSA:

- (1) Very secure.
- (2) Supports digital signatures
- (3) Solves key-distribution problems
- (4) Widely used in Internet

→ Disadvantages of RSA:

- (1) Slower than symmetric algo.
- (2) Requires larger key sizes.
- (3) Computationally expensive.

→ Applications of RSA:

- (1) HTTPS / SSL / TLS
- (2) Secure email
- (3) Digital Certificates
- (4) Digital signatures
- (5) Online banking
- (6) E-commerce security.

→ RSA Algo Example:

① Key Generation:

1. Let's choose two prime numbers, p & q

$$p=3 \quad q=11$$

2. Compute modulo $n = p \times q$

$$= 3 \times 11$$

$$\boxed{n=33}$$

3. Compute Totient $\phi(n) = (p-1) \times (q-1)$

$$= (3-1) \times (11-1)$$

$$= 2 \times 10$$

$$\boxed{\phi(n)=20}$$

④ Choose Public Key (e): $\text{gcd}(e, \phi(n)) = 1$

- We need a number (e) bet^w 1 and 20 that shares no common factors with 20.

— Let's select $e=7$

$$\text{gcd}(7, 20) = 1 \quad \checkmark$$

Public Key Pair $(e, n) = (7, 33)$

5 Calculate Private Exponent (d):

$$(d * e) \equiv 1 \pmod{\phi(n)}$$

$$(d * 7) \pmod{\phi(n)} = 1$$

$$(d * 7) \pmod{20} = 1$$

lets test

$$7 * 3 = 21 \equiv 21 \pmod{20}$$

$$= 21 - 20$$

$$= 1$$

Therefore $d = 3$

Private Key $(d, n) = (3, 33)$

→ Encryption: $C = P^e \pmod{n}$

message = ~~7~~ M (M = 7)

$$C = 7 \pmod{33}$$

$$= \del{823543}$$

$$= 823543 \pmod{33}$$

$$= (24955 * 33) + 28$$

$$= 28 \text{ remainder}$$

Cipher text (C) = 28

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⇒ Decryption $P = c^d \bmod n$

$$P = 28^3 \bmod 33$$

$$= 21952 \bmod 33$$

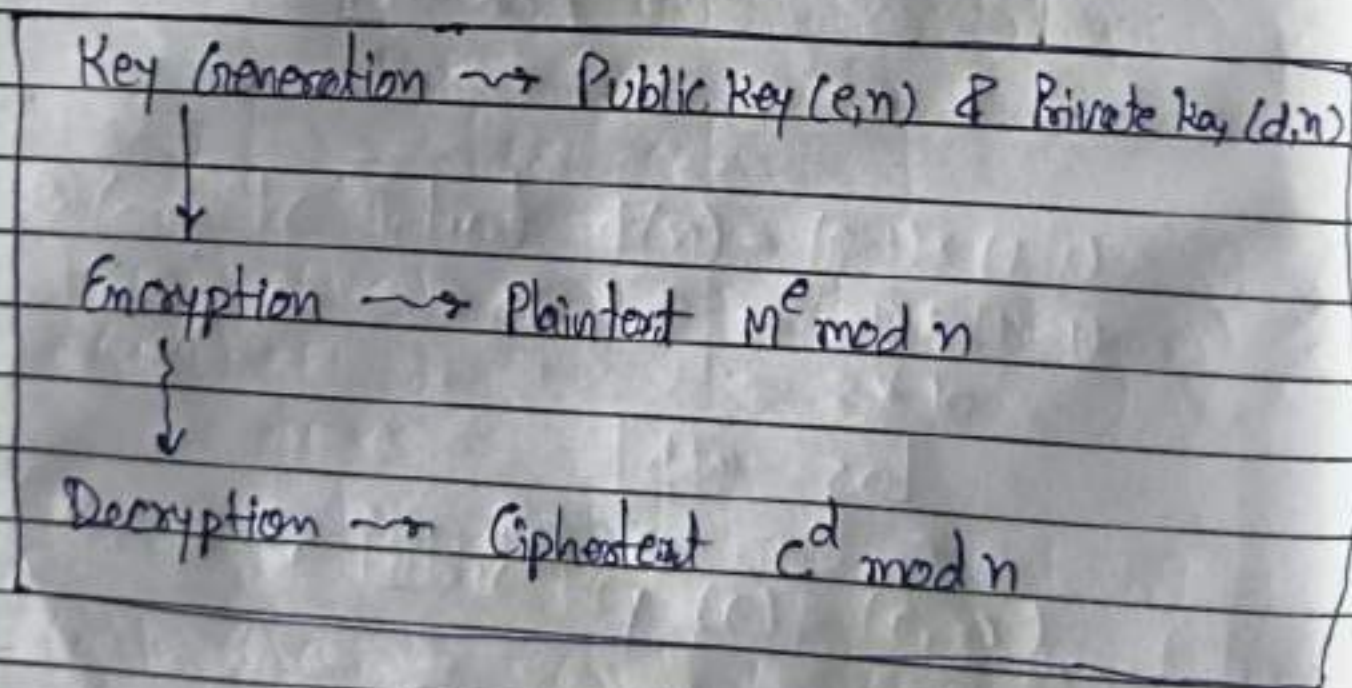
$$= (665 \times 33) + 7$$

$$\boxed{= 7}$$

$$\boxed{\text{Plaintext} = 7 = M}$$

→ The Three Phases of RSA:

- ① The RSA algorithm is an asymmetric cryptographic method that uses a public key for encryption and a private key for decryption.
- ② Three phases:



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② Key Generation:

1. Choose two distinct prime numbers p and q
2. Compute modulo $n = p * q$
3. Compute Euler's Totient $\phi(n) = (p-1) * (q-1)$
4. Select an integer e that meets the following criteria: (Public-Key calculation)

$$1 < e < \phi(n) \quad (\text{not } 1)$$

$$\text{GCD}(e, \phi(n)) = 1 \rightarrow \text{Public Key } (e, n)$$

5. Calculate Private Exponent (d):

$$(d * e) \pmod{\phi(n)} = 1$$

$$\text{Private Key: } (d, n)$$

6. Encryption: $C = m^e \pmod{n}$

7. Decryption: $m = C^d \pmod{n}$

Example: RSA Algo.

① Select two prime p & q

lets $p = 3$ & $q = 11$

② Compute $n = p \times q$
 $= 3 \times 11$

$$n = 33$$

③ Compute Totient $\phi(n) = (p-1) \times (q-1)$

$$= (3-1) \times (11-1)$$

$$= 2 \times 10$$

$$\phi(n) = 20$$

④ Choose Public Key $(e) = (e, n)$

$$\text{GCD}(e, \phi(n)) = 1$$

$$\text{GCD}(e, 20) = 1$$

Select $1 \times 1 = 1$ $20 \Rightarrow 1 \times 2 + 2 \times 5 = 20$ X

$$2 \times 1 = 2$$

$$3 \times 1 = 3$$

$$4 \times 1 = 4$$

$$5 \times 1 = 5$$

$$1 \times 2 + 2 \times 5 = 20$$

is not co-prime

$$6 \times 1 = 6$$

$$7 \times 1 = 7$$

$$e = 3$$

5) Compute Private Key $d = (d, n)$

$$(a \times e) \bmod \phi(n) \equiv 1$$

$$(e \times d) \bmod \phi(n) \equiv 1$$

$$3 \times d \bmod 20 \equiv 1$$

Choice: 1

$$3 \times 1 \bmod 20 =$$

$$3 \bmod 20 \neq 1$$

$$\begin{array}{r} 3 \overline{) 20} \\ \underline{18} \\ 2 \end{array}$$

Choice: 2

$$3 \times 2 \bmod 20 \equiv 1$$

$$6 \bmod 20 \equiv 1 \neq 1$$

$$\begin{array}{r} 3 \overline{) 20} \\ \underline{18} \\ 2 \end{array}$$

Choice: 3

$$3 \times 3 \bmod 20$$

$$9 \bmod 20 \neq 1$$

$$\begin{array}{r} 3 \overline{) 20} \\ \underline{18} \\ 2 \end{array}$$

Choice: 4

$$3 \times 4 \bmod 20$$

$$12 \bmod 20 \neq 1$$

$$\begin{array}{r} 3 \overline{) 20} \\ \underline{12} \\ 8 \end{array}$$

Choice: 7

$$= 7 \times 3 \bmod 20$$

$$= 21 \bmod 20$$

$$= 21 - 20$$

$$\boxed{\equiv 1}$$

$$\begin{array}{r} 7 \overline{) 20} \\ \underline{14} \\ 6 \end{array}$$

$$= (20 \times 1) + 1$$

Reminder

$$\boxed{d = 7}$$

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(6) Encryption $C = P^e \text{ mod } n$

e = public-key
 P = plaintext

Let's Message / Plaintext = B ($B=2$)

$$\begin{aligned} C &= 2^3 \text{ mod } 33 & (e=3) \\ &= 8 \text{ mod } 33 & n=33 \\ &= 8 \text{ (remainder)} \end{aligned}$$

$$C=8$$

Ciphertext $C=8$

(7) Decryption $P = C^d \text{ mod } n$

d = private key

$$\begin{aligned} P &= 8^7 \text{ mod } 33 & (C=8) \\ &= 2 & d=7 \\ & \text{(positive remainder)} & n=33 \end{aligned}$$

$P=2$

Plaintext

→ Example: 3 $p=3$ $q=11$

(1) $p=3$ $q=11$

(2) $n=p \times q \rightarrow \boxed{n=33}$

(3) $\phi(n) = (p-1)(q-1)$ $\phi(n)=20$

⊕ Choose Public-Key $e = (e, n)$

$\text{GCD}(e, \phi(n)) = 1$

$\text{GCD}(e, 20) = 1$

Select (1) $1 \times 1 = 1$ ✗ $20 \neq 1 \times 2 \times 2 \times 5$

(2) $2 \times 1 = 2$ ✗

(3) $3 \times 1 = 3$ ✓

$e=7$

(7) $7 \times 1 = 7$ ✓

(5) Private-Key $d = (d, n)$

$(d \times e) \bmod \phi(n) = 1$

$(d \times 7) \bmod 20 = 1$

Select (1) $1 \times 7 \bmod 20 \neq 1$

(2) $2 \times 7 \bmod 20 \neq 1$

(3) $3 \times 7 \bmod 20$

$= (20 \times 1) + 1$

$= 1$ ✓

$d=3$

→ Encryption $C = P^e \text{ mod } n$

$$n = 33$$

Let's message = $P = 2$

$$e = 7$$

$$C = 2^7 \text{ mod } 33$$

$$= 128 \text{ mod } 33$$

$$= (33 \times 3) + 29$$

$$\boxed{C = 29} \text{ — cipher text}$$

→ Decryption $P = C^d \text{ mod } n$

$$P = 29^3 \text{ mod } 33$$

$$d = 3$$

$$= 24389 \text{ mod } 33$$

$$C = 29$$

$$= (739 \times 33) + 2$$

$$n = 33$$

$$= 2 \text{ (Remainder)}$$

$$\boxed{P = 2}$$

Plaintext

$$24389 \div 33$$

$$\approx 739$$